

Cost-Effectiveness of Reducing Glycaemic Episodes Through Community Pharmacy Management of Patients with Type 2 Diabetes Mellitus

Delia Hendrie · Ted R. Miller ·
Richard J. Woodman · Kreshnik Hoti ·
Jeff Hughes

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Abstract Accessibility, availability and frequent public contact place community pharmacists in an ideal position to provide medically necessary, intensive health education and preventive health services to diabetes patients, thus reducing physician burden. We assessed the cost-effectiveness of reducing glycaemic episodes in patients with type 2 diabetes mellitus through a pharmacist-led Diabetes Management Education Program (DMEP) compared to standard care. We recruited eight metropolitan community pharmacies in Perth, Western Australia for the study. We paired them based on geographical location and the socioeconomic status of the population served, and then randomly selected one pharmacy in each pair to be in

the intervention group, with the other assigned to the control group. We conducted an incremental cost-effectiveness analysis to compare the costs and effectiveness of DMEP with standard pharmacy care. Cost per patient of implementing DMEP was AU\$394 (US\$356) for the 6-month intervention period. Significantly greater reductions in number of hyperglycaemic and hypoglycaemic episodes occurred in the intervention relative to the control group [*OR* 0.34 (95 % CI 0.22, 0.52), *p* = 0.001; *OR* 0.54 (95 % CI 0.34, 0.86), *p* = 0.009], respectively, with a net reduction of 1.86 days with glycaemic episodes per patient per month. The cost-effectiveness of DMEP relative to standard pharmacy care was AU\$43 (US\$39) per day of glycaemic symptoms avoided. Patients with type 2 diabetes in three surveys were willing to pay an average of 1.9 times that amount to avoid a hypoglycaemic day. We conclude that DMEP decreased days with glycaemic symptoms at a reasonable cost. If a larger-scale replication study confirms these findings, widespread adoption of this approach would improve diabetes health without burdening general practitioners.

Keywords Block-randomised · Health education · Indicated prevention · SF36 · Incremental cost-effectiveness ratio

D. Hendrie · T. R. Miller
Centre for Population Health Research, Curtin University,
GPO Box U1987, Perth, WA 6845, Australia
e-mail: D.V.Hendrie@curtin.edu.au

T. R. Miller (✉)
Pacific Institute for Research and Evaluation, 11720
Beltsville Drive, Suite 900, Calverton, MD 20705, USA
e-mail: miller@pire.org

R. J. Woodman · K. Hoti · J. Hughes
School of Public Health, Curtin University,
GPO Box U1987, Perth, WA 6845, Australia
e-mail: Richard.Woodman@flinders.edu.au

K. Hoti
e-mail: Kreshnik.Hoti@curtin.edu.au

J. Hughes
e-mail: J.D.Hughes@curtin.edu.au

Diabetes contributes significantly to premature mortality, morbidity, disability and lost work time (American Diabetes Association, 2008; Australian Institute

of Health and Welfare, 2008, 2011). Type 2 diabetes mellitus accounts for more than 95 % of diabetes globally, and its rate is increasing faster than type 1 diabetes (Amos, McCarty, & Zimmet, 1997; Day, 2001). The rising rate of type 2 diabetes and the intense management required to support cases raise concerns about the Australian health care system's capacity to afford type 2 diabetes management delivered solely by general practitioners (Krass et al., 2007). An alternative strategy is a team approach that assigns community pharmacists responsibility for components of disease management. Given their accessibility, availability and frequent public contact, community pharmacists are ideally placed to provide medically necessary health education and preventive health services to patients with chronic conditions (Mitchell et al., 2011).

Meta-analysts report that current evidence related to the benefits of expanding the role of community pharmacists is inconclusive (Mossialos, Naci, & Courtin, 2013). However, studies examining effects of pharmacists on diabetes management teams generally have shown positive outcomes. Changes have included clinically meaningful improvements in glycaemic control (Machado, Bajcar, Guzzo, & Einarson, 2005), gains in self-reported quality of life (Krass & Dhippayom, 2013), and improved patient knowledge regarding diabetes management (Al Mazroui, Kamal, Ghabash, Yacout, Kole, & McElnay, 2009; Kelly & Rodgers, 2000; Rothman, Malone, Bryant, Horlen, & Pignone, 2003; Taylor, Milanova, Hourihan, Krass, Coleman, & Armour, 2005). However, almost no literature has examined the economics of pharmacist-led diabetes programs focused on the improvement of primary care outcomes of patients with diabetes. A US study evaluated the cost of an operational diabetes program provided by three clinical pharmacists and a case manager. The intervention consisted of monthly sessions of education and drug therapy management. It cost US\$37 per patient per month (AU\$49) with a marginal labour cost of program implementation of US\$25 per patient per month (AU\$33; Rothman, So, & Shin, 2006). An Australian cost-effectiveness study of pharmacist-led medication review and education reported a cost of AU\$383 (US\$346) per patient to achieve a 0.43 % reduction in haemoglobin A1c (HbA1c) over 9 months (Taylor et al., 2005). An economic analysis of the Diabetes Medication Assistance Service in Australia indicated that, over a

6 month period, the variable cost per patient was AU\$272 (US\$246; Krass, Armour, Clarke, Peterson, Stewart, & Hughes, 2005).

We conducted the Diabetes Mellitus Education Program (DMEP) in Perth, Western Australia. DMEP involved community pharmacists providing customised diabetes education sessions and ongoing follow-up support to patients with type 2 diabetes. The aims of the program were to improve patients' knowledge and understanding of diabetes and its management, and in doing so, to improve their metabolic control (Hughes, Tenni, Coles, & Wilkinson, 2006).

Given pressures on health care resources, it is important to examine not only if DMEP achieved its aims, but whether it achieved them at reasonable cost. This paper presents an evaluation and cost-effectiveness analysis of DMEP that compares its costs and outcomes. We measured DMEP effectiveness based on frequency of hyperglycaemic and hypoglycaemic episodes at study exit relative to episodes at initial visit, as well as self-reported knowledge and functional gains. We believe this is the first study focusing on cost-effectiveness of reducing glycaemic episodes by a pharmacist-led diabetes program.

Methods

Study Design

We recruited eight metropolitan community pharmacies in Perth, Western Australia for the study. We paired them based on geographical location and the socioeconomic status of the population served, and then randomly selected one pharmacy in each pair to be in the intervention (DMEP protocol) group, with the other assigned to the control (standard care) group. The Curtin University Human Research and Ethics Committee approved the study.

The project pharmacist at each pharmacy attended an 8-h training day covering areas such as patient recruitment and interview format. Three education pharmacists delivered the DMEP intervention. They attended training with the project pharmacists and eight additional hours of intensive training by an accredited diabetes educator. This training focused on techniques for communicating complex concepts such as the mode of action of anti-diabetic medications. It also included hands-on training with the software used

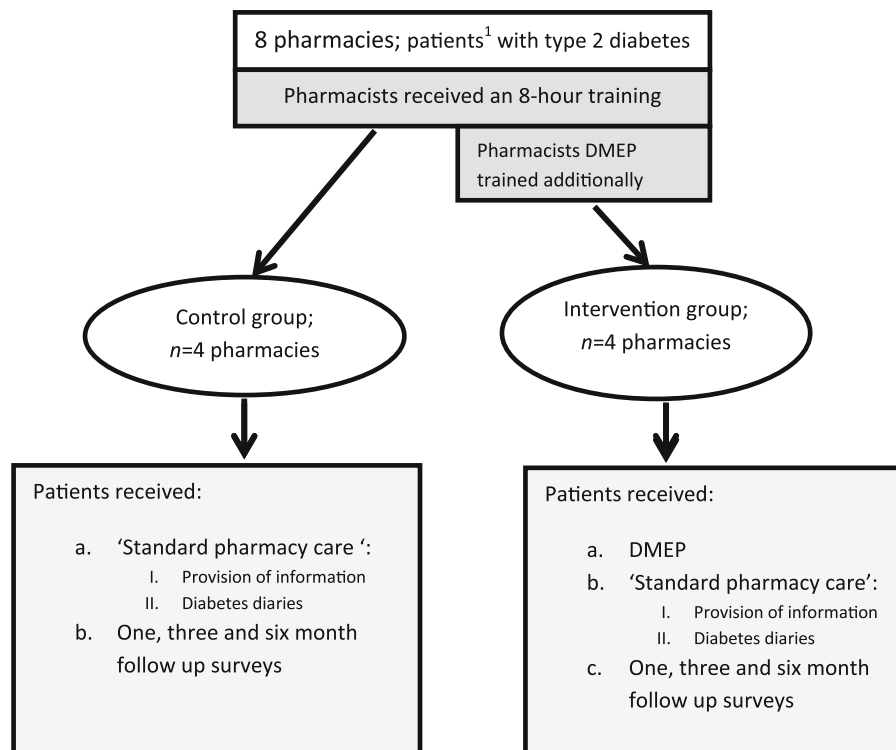


Fig. 1 Outline of the study protocol. This figure shows the flow of participants through the study. *DMEP* Diabetes Management Education Program. ¹Exclusion of patients under the age of 18,

pregnant women, patients from non-English speaking backgrounds, those with intellectual or mental impairment, and those highly dependent on medical care

to customise each patient's diabetes education program.

Study Population

The study population comprised patients aged 18 years or older with type 2 diabetes. Pharmacists recruited patients when they presented a prescription for diabetes test strips or an anti-diabetic medication (an oral hypoglycaemic agent or insulin). The study excluded pregnant females, patients from non-English speaking backgrounds, those with intellectual or mental impairment, and those highly dependent on medical care such as the chronically disabled. Patient enrolment occurred between May 2003 and August 2004.

Intervention

Patients in the intervention group received DMEP, a tailored education program delivered by one of the

trained pharmacist diabetes educators. Patients completed a modified version of the Diabetes Patient Assessment Questionnaire (DPAQ; Campbell & Bennett, 2002). Their pharmacist-educator entered the DPAQ responses into the diabetes disease management module of CogniCare[®], a Windows-based pharmaceutical care software program designed for point-of-care cognitive service provision (not referenced as the vendor no longer exists). The module helped the pharmacist to tailor an individualised program containing up to 3 h of one-on-one diabetes education. In collaboration with the pharmacist, patients established personal treatment targets for the following six-month period. Pharmacists also provided patient education materials, primarily fact sheets from the International Diabetes Institute, Diabetes Australia, Pharmacy Self-Care, and the National Heart Foundation. Following completion of the education sessions, the pharmacist followed up with patients at 1, 3 and 6 months. Activities at these visits, in addition to standard pharmacy care, included

completion of Progress Check List Questionnaires, outcome monitoring, further education, goal review, and identification and resolution of barriers to program adherence. Figure 1 summarizes the study design.

Standard Care

Patients in the control group received standard pharmacy care, which for the purposes of this study included the following components. At an initial visit, the pharmacist presented them with a diabetes diary that included written information on diabetes and its management, their medication regimen, and dates for prescription refills, follow-up visits, and general practitioners' appointments. They asked patients to record data in their diaries detailing diabetes complications and their management, use of health services, and other related information. At the 1, 3 and 6 month follow up, patients completed a Progress Check List Questionnaire, and pharmacists reviewed their diaries and delivered standard pharmacy care.

Data Collection

We collected data in patient diaries, at pharmacy visits, and from each patient's general practitioner. At each pharmacy visit, Questionnaire about control of their diabetes and their understanding of its management. In addition, at enrolment and the 6-month (study exit) visit, patients completed a modified version of the DPAQ and the Diabetes Symptom Checklist (Grootenhuis, Snoek, Heine, & Bouter, 1994), as well as the Short Form 36 (SF36) functional status instrument (Ware, Kosinski, Bayliss, McHorney, Rogers, & Raczek, 1995). We obtained available clinical data (e.g., blood pressure, HbA1c, lipids) from the patient's general practitioner at the same time intervals, but made no attempt to influence what tests physicians ordered. For example, only 56 % of DMEP and 40 % of control patients had HbA1c tested both pre and post intervention. Since these patients probably are not representative of the broader patient group, we could not use their data to analyse DMEP's effects on HbA1c levels.

Economic Analysis Study Type

We performed an incremental cost-effectiveness analysis, with costs and effectiveness of DMEP compared

with standard pharmacy care. We used the health sector's perspective, which means that we counted only costs to the health care system of implementing DMEP and excluded costs to others (notably the value of the time participants spent and any travel costs they incurred).

Our analyses included all additional direct costs of providing DMEP, as if it were implemented as an ongoing program. These consisted of training costs of participating pharmacists and costs of developing, implementing, and operating the customised education program. We calculated costs directly from data collected during the study.

Training costs were funding provided to employ relief pharmacists to cover for project and educator pharmacists while they attended training days. Following training, we assumed pharmacists could provide diabetes management education care for 3 years before requiring more training. Since DMEP completed all interventions within 7 months after training, we allocated 7/12 of annualised training cost to DMEP.

Development, implementation, and operating costs included costs of customising patient plans, educational materials, and education delivery. We costed laptop computers and software used by educator pharmacists to individualise patient plans with a 3-year useful life. Of the annualized cost, we allocated 7/12 to DMEP. We based the cost of developing individualised diabetes education on the estimated time taken by educator pharmacists to prepare customised programs.

Since patient education materials were existing brochures in the public domain, we included only printing costs for these materials. We excluded the development costs of patient diary templates, since this was a one-time cost. The diary template will be available for free in replication. Again, we included only the cost of printing diaries.

Program delivery included one-on-one sessions delivered by educator pharmacists, initial and follow-up visits with project pharmacists, and reminder calls to patients the day prior to their appointment. We included the cost of the initial and follow-up visits, even though the control group also had these sessions, as the visits were not part of standard pharmacy care. We costed education sessions based on time spent by educator pharmacists with patients and the uniform statewide reimbursement rate per hour of pharmacists'

counselling time. We added flat fees paid to pharmacies for initial and follow-up visits. We costed telephone calls based on the metropolitan call charge.

We inflated all program costs to 2011 Australian dollars using average weekly earnings for pharmacist time and the general consumer price index for all other costs.

Measure of Effectiveness

We measured program effectiveness by changes in patients' number of days with symptoms of hyperglycaemia and hypoglycaemia at study exit relative to study entry. We obtained these data from the self-reported responses to questions in the modified DPAQ asking (1) how many days in the last month they had high blood sugar with symptoms such as thirst, dry mouth and skin, increased sugar in the urine, less appetite, nausea, or fatigue and (2) how many times in the last month they had low blood sugar reaction with symptoms such as sweating, weakness, anxiety, trembling, hunger, or headache. We assumed that hypoglycaemic symptoms occur only once a day, so the number of times patients reported low blood sugar reactions was equivalent to the number of days with low blood sugar reactions. At study exit, treatment and control respondents should have been able to use their diaries to recall with accuracy the number of days they experienced symptoms of hyperglycaemia and hypoglycaemia.

The modified DPAQ specified the number of occasions with high or low blood sugar levels as 0, 1–3, 4–6, 7–12 days, and more than 12 days. We assumed a mid-point value for each class interval, and a value of 15 for the more than 12 days interval. This enabled our calculation of the average number of days in the last month patients experienced symptoms of hyperglycaemia and hypoglycaemia for case and control groups at both baseline and study exit. However, we used the categorical responses to test for a statistically significant difference in number of days with symptoms of hyperglycaemia and hypoglycaemia at study exit relative to the initial visit.

We measured DMEP effectiveness as changes in patients' number of days with glycaemic episodes at baseline versus 6 months for the intervention relative to the control group members. Reduced glycaemic episodes indicates better control of blood sugar levels, and studies show it improves health-related quality of life (Kleefstra, Ubink-Veltmaat, Houweling, Groenier,

Meyboom-de Jong, & Bilo, 2005; Lundkvist, Berne, Bolinder, & Jonsson, 2005). Moreover, recent clinical trials and resulting changes in diabetes management guidelines suggest reducing hypoglycaemic episodes may be more critical to survival than tight glycaemic control (American Diabetes Association, 2013; Aro-now, 2013).

Incremental Cost-Effectiveness Ratio

We calculated the incremental cost-effectiveness ratio (ICER) as additional costs to the health sector to provide DMEP divided by the reduction in patients' number of days with glycaemic episodes. Patients in a US internet survey rated the benefits of reducing hyperglycaemic and hypoglycaemic episodes as roughly equal (Peyrot & Rubin, 2011). To judge cost-effectiveness, therefore, we compared the cost per glycaemic episode avoided to the average estimated willingness to pay to reduce hypoglycaemic episodes across three surveys: European Union data from 2001 (Aristides, Weston, FitzGerald, Le Reun, & Maniadakis, 2004), United Kingdom data from 2005 to 2006 (Lloyd et al., 2011), and Canadian data from 2008 (Guimarães et al., 2009) (personal communication, L. Lynd, May 2013). We converted willingness to pay to Australian dollars using foreign exchange rates, and then inflated to 2011 dollars using the Australian Consumer Price Index.

Sensitivity Analysis

We tested the robustness of study results with sensitivity analyses that calculated 'worst -case' and 'best-case' scenarios by varying the magnitude of selected parameters. The worst-case scenario increased time spent by educator pharmacists with patients in the intervention group by one standard deviation. The best-case scenario assumed that pharmacists only require retraining in diabetes management education every 5 years, and a mid-point value of 25 days for the class interval of more than 12 days that patients experienced hyperglycaemic and hypoglycaemic episodes. An alternative scenario set varied pharmacist salaries by $\pm 20\%$.

Statistical Procedures

We analysed study data using SPSS 12.0TM and STATA version 9.0 (StataCorp, Texas). We tested the

significance of group differences at baseline using contingency tables. We included costs for all patients who began the study and assumed zero program effectiveness for patients who did not complete the study (an intent to treat approach). We used ordinal logistical regression based on the proportional odds assumption to test for statistically significant differences between the intervention and control groups in several process outcomes and in patients' number of days with glycaemic episodes at 6 months with adjustment for the number at baseline. We calculated an odds ratio of the likelihood that patients in the intervention group relative to the control group would be in a higher (i.e. worse) glycaemic episode frequency category at 6 months than at baseline. We assessed clinical significance of changes in SF36 domain scores using a one-standard error of measurement criterion (Wyrwich, Tierney, & Wolinsky, 1999).

Results

Study Enrolment

We randomized 245 patients, who included 119 (49 %) in the intervention group and 126 (51 %) in the control group. Of these, 57 refused enrolment and 28 declined to schedule an initial visit after completing part or all of the enrolment survey. Thus, the study started with just 70 patients in the intervention group and 90 in the control group. Thirteen intervention group patients (18.6 %) and 17 control group patients (18.9 %) dropped out of the study for various reasons.

Baseline Data

The intervention and control groups were well-matched in relation to patients' demographic and diabetes-related characteristics (Table 1). Participants recruited to both intervention and control groups averaged 64 years. Approximately half of each group were female. The body mass index (BMI) of patients in both groups was mostly in the obese category (BMI of 30 or greater). Only six patients—all controls—were not taking diabetes medication at baseline and insulin usage rates were similar in both groups. The groups did not differ significantly either in the duration

Table 1 Baseline characteristics of control and intervention patients

Variable	Control	Intervention	<i>p</i> value
Cases	114	83	
Gender (Male)	47.4 %	45.8 %	0.83
Mean age (Years)	64.3	64.0	0.82
Mean body mass index (Kg/m ²)	30.3	31.2	0.37
Duration of diabetes			
< 1 year	10.9 %	4.9 %	0.40
1–2 years	10.0 %	13.4 %	
3–5 years	20.0 %	19.5 %	
6–10 years	28.2 %	36.6 %	
>10 years	30.9 %	25.6 %	
Days in the last month with symptoms of hyperglycaemia			
Not at all	37.6	37.0 %	
1–3 days	17.4 %	11.1 %	
4–6 days	9.2 %	9.9 %	
7–12 days	9.2 %	6.2 %	0.64
>12 days	12.8 %	14.8 %	
Don't know	13.8 %	21.0 %	
Hypoglycaemic attacks in the last month			
0 times	66.7 %	61.0 %	
1–3 times	18.9 %	20.7 %	
4–6 times	3.6 %	6.1 %	0.95
7–12 times	2.7 %	2.4 %	
>12 times	0.9 %	1.2 %	
Don't know	7.2 %	8.5 %	

Baseline characteristics shown for patients who completed entry questionnaires

The *p* values are based on Chi squared computed from contingency tables

of their diabetes or frequency of their hyperglycaemic or hypoglycaemic attacks in the last month.

Program Cost

As displayed in Table 2, the total cost of DMEP was AU\$27,591 (US\$24,920), or AU\$394 (US\$356) per patient who started the program (\$27,591/70). Consistent with an intent-to-treat approach, we incorporated costs of recruitment and partial program delivery for non-completers into the costs per program completion. Training costs accounted for a small share (4.3 %) of total costs. Major cost items were time

Table 2 Cost of implementing the diabetes management education program

	No. of units	Cost per unit (Au \$)	Total cost (Au \$)
<i>Training</i>			
Training sessions			
Project pharmacists	4	119	476
Educator pharmacists	3	119	357
Intensive diabetes education training			
Educator pharmacists	3	119	357
<i>Intervention implementation</i>			
Development, equipment, and materials			
Laptop computer	3	610	1,830
Cognicare [®] software	3	420	1,260
Powerpoint presentations	3	360	1,080
Handouts/brochures ^a	490	0.22	108
Diaries ^b	140	3	420
Program delivery			
Initial visits ^c	70	41	2,870
Education sessions ^d	70	144	10,080
Telephone calls ^e	210	0.30	63
Follow-up visits ^f	210	41	8,610
Telephone calls ^g	266	0.30	80
Total cost			\$27,591

Per hour costs are based on universal reimbursement rate in Western Australia for accredited pharmacists to deliver cognitive services

^a Mean number of handouts of 7.0 per patient

^b Two diaries per patient

^c ½ h per visit at \$60/h

^d Mean total time of education sessions of 106 min per patient at \$60/h

^e Mean number of telephone calls of 3.0 per patient

^f Three follow up visits per patient, ½ h per session at \$60/h

^g Mean number of telephone calls of 3.8 per patient

spent by educator pharmacists with patients early in the program (36.5 %) and the time project pharmacists spent with patients during the three follow up visits (31.2 %).

Effectiveness

Process outcomes were promising. Participants reported that they felt better about their diabetes care [*OR* 0.51, 95 % confidence interval (CI) 0.19, 1.43, *p* = 0.20] and felt ready to work to improve their

diabetes care [*OR* 0.38, 95 % CI 0.19, 0.76, *p* < 0.01]. More than half of DMEP completers reported knowledge gains about diet and prevention of complications, and more than a quarter reported knowledge gains about exercise, weight management, medication, and monitoring of blood glucose levels. The percentage of DMEP participants with recent HbA1c tests increased significantly relative to controls (*p* < 0.01), reaching 84 %.

In terms of health outcomes, DMEP participants had significantly fewer glycaemic episodes than controls at study exit after adjustment for baseline. For patients' number of days in the last month with hyperglycaemia, the odds ratio from ordinal logistical regression was 0.34 (95 % CI 0.22, 0.52, *p* < 0.01) for the intervention group relative to the control group. This implies a 66 % reduction in odds of being in a higher class interval (more days in the last month with symptoms of hyperglycaemia) relative to a lower class interval. For number of times in the last month with hypoglycaemia, the odds ratio was 0.54 (95 % CI 0.34, 0.86, *p* < 0.01) for the intervention group relative to the control group.

At baseline, patients in the control and intervention groups averaged 3.95 days per month and 3.40 days per month with hyperglycaemic symptoms, respectively, a difference of 0.55 (see Table 3). The difference in the number of days with hyperglycaemic episodes increased to 1.81 days per month at study exit, implying the intervention group had gained 1.26 days per month without hyperglycaemic episodes relative to the control group. In the case of hypoglycaemic episodes, the equivalent gain by the intervention group relative to the control group was 0.60 days per month without symptoms at study exit relative to baseline. Program effectiveness, measured by the reduction in the number of days with symptoms of hyperglycaemia and hypoglycaemia, was thus 1.86 days per patient per month. If the reduction in number of days with glycaemic symptoms occurred each month during the six-month study duration, the commensurate gain per patient would be 11.16 days without glycaemic episodes. Assuming zero effectiveness for patients who did not complete, the gain per patient reduces to 9.09 days without glycaemic episodes.

Relative to controls, DMEP patients reported clinically significant improvements in all four mental and emotional domains of the SF36. With the small

Table 3 Program effectiveness measured as reduction in days with symptoms of hyperglycaemia and hypoglycaemia

	Average number of days per month		
	With hyperglycaemia symptoms		With hyperglycaemic symptoms avoided Study exit
	Baseline	Study exit	
Control group [C]	3.95	2.88	
Intervention group [I]	3.40	1.07	
Difference: [C] – [I]	0.55	1.81	1.26 ^a
	With hypoglycaemic symptoms		With hypoglycaemic symptoms avoided 6 Months follow-up
	Baseline	6 Months follow-up	
Control group [C]	0.79	0.84	
Intervention group [I]	0.97	0.42	
Difference: [C] – [I]	–0.18	0.42	0.60 ^a

^a Represents the change in the difference between the average number of days with glycaemic symptoms for the control and intervention groups at study exit compared with baseline. In the case of the average number of days with hyperglycaemic symptoms avoided, 1.26 equals 1.81 minus 0.55. Similarly, 0.60 equals –0.18 minus 0.42. The text describes how mean days were inferred from five categorical responses (e.g., 7–12 days)

sample of completers, however, only the gain in Social Functioning was statistically significant ($p = 0.01$), with regression-adjusted significance levels for the other domains ranging from 0.09 to 0.28.

Incremental Cost-Effectiveness Ratio

The ICER of DMEP relative to standard pharmacy care was the cost of its implementation divided by its effectiveness. It amounted to AU\$43 (US\$39) per day of glycaemic symptoms avoided.

Sensitivity Analysis

The assumption in the ‘worst-case’ scenario that time spent by educator pharmacists with patients was one standard deviation (44 min) above the mean value (106 min) increased total cost of the DMEP to AU\$31,721 (US\$28,650), its mean cost per patient starting the program to AU\$453 (US\$409), and the ICER to AU\$50 (US\$45) per day of glycaemic episodes avoided.

In the best-case scenario, if pharmacists needed retraining in diabetes management education every five years rather than every three years, and the mid-point value increased to 25 from 15 for the class interval of more than 12 days for the number of days with hyperglycaemic and hypoglycaemic episodes, total cost

of the DMEP decreased to AU\$27,141 (US\$24,513), mean cost per patient to AU\$388 (US\$350), and the ICER to \$34 (US\$31) per day of glycaemic symptoms avoided. Assuming that mean pharmacist salaries in implementation shifted by $\pm 20\%$ from Western Australia’s fixed reimbursement levels would have yielded a wider ICER range of AU\$36 to \$51 (US\$33 to \$46) per day of glycaemic episodes avoided.

Discussion

In this study, a customised diabetes education program administered by pharmacists to patients with type 2 diabetes decreased the days they experienced glycaemic symptoms by 1.86 per month in the intervention group relative to a control group who received standard pharmacy care. This health gain cost AU\$43 (US\$39) per day of glycaemic symptoms avoided. These findings should be viewed in the context of leveraging accessible community pharmacies and pharmacists to deliver an intervention that could improve diabetes control and be cost-effective. Our study’s findings are particularly relevant because a number of interventions have proven ability to reduce the cost burden of diabetes but resources to deliver them are limited (Li, Zhang, Barker, Chowdhury, & Zhang, 2010).

People with type 2 diabetes in three surveys covering Canada, France, Germany, Italy, Spain, and the United Kingdom were willing to pay an average of AU\$81 (US\$73) to avoid a hypoglycaemic day, or 1.9 times DMEP's AU\$43 (US\$39) cost per day of glycaemic symptoms averted. Assuming a similar willingness to pay for hyperglycaemia and for Australia (an assumption that we make in an empirical vacuum), DMEP would be a sound expenditure.

Hypoglycaemia and hyperglycaemia reduce patient well-being not only because of acute symptoms but also from behaviour changes in response to fear of glycaemic episodes. Hypoglycaemia has the potential to cause rebound hyperglycaemia, imposing further acute symptoms. On the other hand, hyperglycaemia raises the risk of diabetic complications. DMEP resulted in improved glycaemic episodes, which implies a positive effect on normoglycaemia. Benefits of achieving normoglycaemia in patients with diabetes are well-established (The Diabetes Control and Complications Trial Research Group, 1993). Eastman et al. (1993) concluded that the cost-effectiveness of treating type 2 diabetes with the goal of achieving normoglycaemia is comparable to other cost-effective interventions. An economic analysis by the United Kingdom's prospective diabetes study showed that the added costs of intensively managing diabetes are compensated by cost reductions in treating diabetes complications (Gray et al., 2000).

Our literature review revealed that health-related quality of life falls as the frequency and severity of hyperglycaemic and hypoglycaemic episodes increase. Goddijn, Bilo, Feskens, Groeniert, van der Zee and Meyboom-de Jong (1999) found that hyperglycaemic episodes were associated with lower scores on the SF36 physical functioning, vitality, and health change domains. Kleefstra et al. (2005) found a larger number of hyperglycaemic symptoms were associated with lower scores on all SF36 domains, as well as on physical and mental component summary scores. Davis, Morrissey, Peters, Wittrup-Jensen, Kennedy-Martin and Currie (2005) found that EQ-5D utility scores and scores on all eight domains of the SF36 declined as hypoglycaemia severity rose. These investigators also found that hypoglycaemia increased days lost from work and primary care resource use. Our current study echoes these findings, with a reduction in glycaemic episodes associated with increased SF36 scores in the four mental/emotional domains.

Limitations

This study has several limitations. First, it was a small trial with far fewer patients recruited than the target of 200 in each of the intervention and control groups. This smaller sample suggests that this study should be viewed as a pilot. Second, the attrition rate was substantial, with 35 % of patients randomized into the study deciding not to participate. Among participants, 19 % did not complete the 6-month exit study. If treatment group dropouts saw some health improvement, the ICER would improve. Third, the educational materials provided to both control and treatment groups in the form of the patient diaries and the requirement for all patients to complete a Progress Check List Questionnaire at each follow-up session had the potential themselves to improve patient knowledge and change patient behaviour. If that occurred, the analysis would have underestimated the effectiveness of the DMEP. Fourth, we based our effectiveness measure on the number of days patients reported glycaemic episodes. We asked all patients to complete diaries, and those who complied may have replaced recall with records at study exit, but at baseline would have relied on recall. Furthermore, if DMEP patients kept diaries more diligently than control group patients, their self-reports may have been more accurate, which could bias comparisons between groups. Fifth, our effectiveness measure, while demonstrating an association with health-related quality of life (Davis et al., 2005; Goddijn et al., 1999; Kleefstra et al., 2005), was far from ideal. Other cost-effectiveness studies of diabetes management programs have not used it, so the ICER for providing customised education program versus standard pharmacy care cannot be compared to the results of ICERs from the other studies. Finally, although we collected only six months of outcome data, outcome improvements may well have continued, which would improve DMEP's cost-effectiveness.

Conclusion

In conclusion, the DMEP intervention that pharmacists provided significantly reduced the number of glycaemic events in the intervention group, which translated into improved health-related quality of life. Implementing DMEP cost AU\$394 (US\$356) per

patient. The program appeared cost-effective with AU\$43 (US\$39) needed to avert a glycaemic episode. However, longer-term sustainability of reducing glycaemic events after program cessation is unknown.

Community pharmacists constitute ideal providers of prevention-focused components of diabetes management. The study demonstrated the success of implementing a diabetes education program in community pharmacies that effectively reduced glycaemic events at reasonable cost. Pharmacists' management of care of patients with chronic diseases such as type 2 diabetes has the potential to improve patient care outcomes.

A larger-scale replication study is warranted. Confirming the findings would suggest that widespread government-reimbursed adoption of this approach would improve diabetes health without burdening general practitioners.

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Conflict of interest The authors have no conflicts of interest.

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